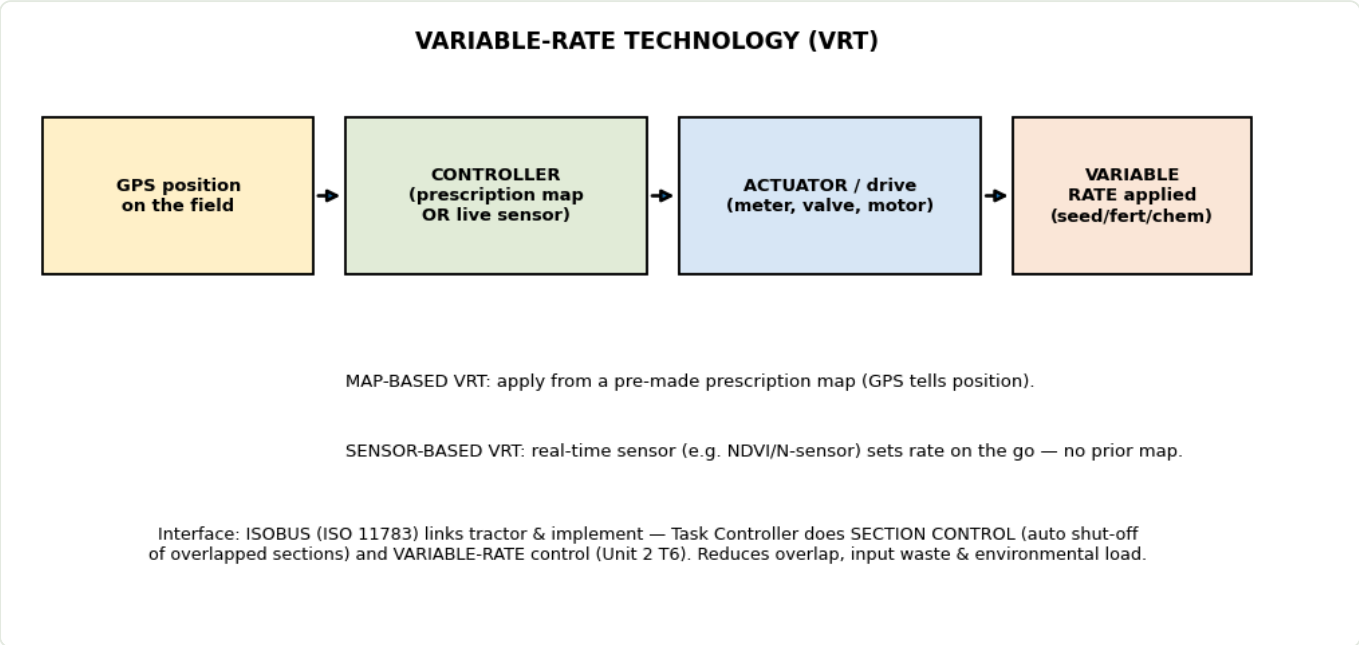
UNIT 10 · TOPIC 9

Electronics and Control for Variable-Rate Application

Syllabus topic: Electronics and control systems for variable rate applications; precision variable equipment



Variable-rate technology (VRT) — control chain and the two approaches

Two approaches to VRT

Approach	How the rate is set
MAP-BASED VRT	The controller reads a pre-made PRESCRIPTION MAP and, using the GPS position (Topic 3), sets the rate for each location
SENSOR-BASED VRT	A real-time canopy/soil sensor sets the rate on the go — no prior map needed

The control chain

1. GPS position or live sensor reading.
2. CONTROLLER (ECU).
3. ACTUATOR — variable-speed metering drive, control valve or motor.
4. Applied rate.

- Sensors, ECUs and actuators form closed loops (mechatronics — Unit 2, Topic 10).
- Centimetre-accurate RTK guidance (Topic 3b) keeps passes precise.

Precision variable equipment

- Variable-rate seed drills/planters — change seed rate/spacing.
- Fertiliser applicators — spinner/pneumatic with variable metering.
- Sprayers — rate, nozzle and section control (Topic 10).

Practice MCQs

Q57. Map-based VRT applies inputs using:

(a) A real-time sensor only
(b) A pre-made prescription map + GPS position
(c) Manual judgement
(d) Fixed rate

Q58. Sensor-based VRT differs in that it:

(a) Needs a prior map
(b) Sets the rate in real time from a live sensor, no prior map
(c) Uses no electronics
(d) Is manual

Q59. The VRT control chain is:

(a) Actuator → GPS → sensor
(b) GPS/sensor → controller → actuator → applied rate
(c) Load cell → engine
(d) Map → hand

Q65. Centimetre-accurate guidance for VRT/auto-steer needs:

(a) Autonomous GPS
(b) RTK correction
(c) A compass
(d) No GPS

Answers

Q57 — (b) A pre-made prescription map + GPS position — *Map-based VRT = prescription map + GPS.*

Q58 — (b) Sets the rate in real time from a live sensor, no prior map — *Sensor-based = real-time sensor, no prior map.*

Q59 — (b) GPS/sensor → controller → actuator → applied rate

Q65 — (b) RTK correction — *RTK gives centimetre accuracy.*

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.